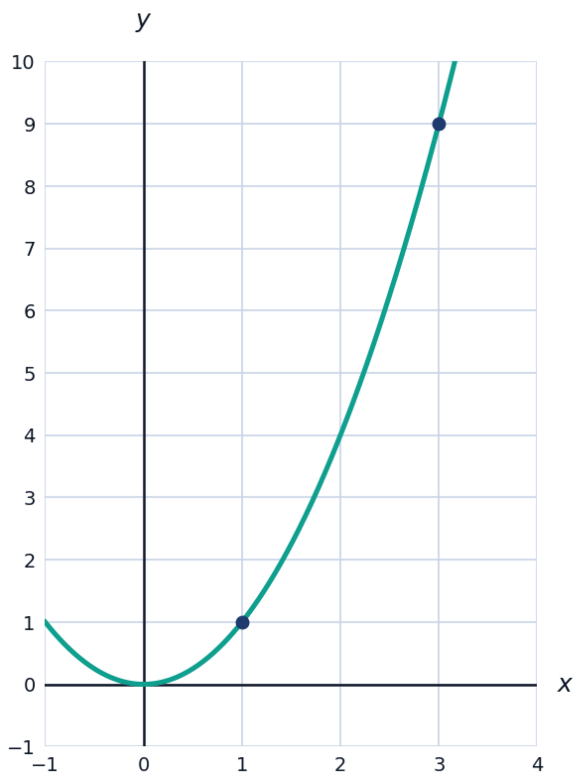
The Derivative from First Principles

The definition of the derivative

1 Use the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$ for $f(x) = x^2 + 3x$.

2 Use the limit definition to find $f'(x)$ for $f(x) = \sqrt{x}$ (for $x > 0$).

3 The graph shows $f(x) = x^2$ with the secant line through $(1, 1)$ and $(3, 9)$.



a Find the slope of the secant line.

b Find $f'(2)$ using the derivative, and compare it to the secant slope.

Differentiability

4 Explain why $f(x) = |x|$ is not differentiable at $x = 0$, using one-sided derivatives.

5 State the relationship between differentiability and continuity: if a function is differentiable at a point, must it be continuous there? Is the converse true?

More derivatives from the definition

6 Use the limit definition to find $f'(x)$ for $f(x) = \frac{1}{x}$ (for $x \neq 0$).

7 Use the limit definition to find $f'(x)$ for $f(x) = x^3$.

Interpreting the derivative

8 A ball's height is $h(t) = -5t^2 + 20t + 2$ (metres, seconds). Use the derivative definition idea to explain what $h'(2)$ represents physically, and find its value using $h'(t) = -10t + 20$.

Point-derivatives from the definition

9 Use the limit definition $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$ to find $f'(4)$ for $f(x) = x^2 - 3x$.

10 Use the alternate form $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$ to find $f'(2)$ for $f(x) = x^3 - 1$.

The derivative as the slope of the tangent line

11 For $f(x) = x^2 - 4x + 1$, use the derivative (found via the limit definition, $f'(x) = 2x - 4$) to find the equation of the tangent line at $x = 3$.

12 Explain why the tangent line to $y = x^2$ at $x = 0$ is horizontal, using the derivative $f'(x) = 2x$ found from the limit definition.

The derivative function versus the derivative at a point

13 Explain the difference between $f'(x)$ and $f'(3)$, using $f(x) = x^2$ as an example.

Estimating a derivative from a table

14 A table gives $f(1.9) = 8.61$, $f(2.0) = 9.00$, $f(2.1) = 9.41$. Use a symmetric difference quotient to estimate $f'(2)$.

The derivative of a constant and linear functions

15 Use the limit definition to show that the derivative of a constant function $f(x) = c$ is 0.

16 Use the limit definition to show that the derivative of a linear function $f(x) = mx + b$ is the constant m.